Energy-Efficient Handover in Multi-Beam LEO Satellite Networks: A Multi-Objective Reinforcement Learning Approach with Catfish Coordination

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***Abstract*—Handover decision-making is a central problem in multi-beam Low Earth Orbit (LEO) satellite networks, where satellite mobility causes frequent, capacity-constrained beam reassignments. The Multi-Objective Deep Q-Network (MODQN) casts it as multi-objective reinforcement learning over throughput, handover cost, and load balancing, but has two limitations: its throughput reward ignores the angle- and power-dependence of energy efficiency (EE), and its per-user argmax lets similarly situated users converge on one beam, starving tail users under a per-satellite active-beam budget. This paper proposes Multi-Catfish Coordinated Reinforcement Learning (MCCRL): the throughput reward becomes an angle-aware EE objective, and the online decision is split into a valuation step of three per-objective catfish roles and a coordinated beam allocation replacing the per-user argmax. Under a tight beam budget, MCCRL raises the served fraction from 0.29 to 0.97–0.99 and the minimum user coverage from 0 to 0.999, beating plain MODQN and the rule-based static baselines (RSS-max, round-robin) on the normalized weighted metric with separated confidence intervals. An ablation attributes the de-concentration to the coordinated allocation step; a fixed-rule coordinated allocation captures most of the equity gain, while learning lowers the handover cost. The advantage is located on the efficiency–equity axis: MCCRL is the only learned method that simultaneously attains high EE and near-full coverage at tight capacity, whereas a scalar-trained DQN attains a higher weighted score only by starving the tail.**

***Index Terms*—beam management, energy efficiency, handover, LEO satellite networks, multi-objective reinforcement learning**

I. Introduction

Low Earth Orbit (LEO) satellite networks are a key enabler of future non-terrestrial networks, offering wide-area coverage, low propagation delay, and flexible deployment [1]. Because LEO satellites move at high speed relative to the ground, the user–satellite–beam association changes rapidly and handover becomes frequent, a central problem in multi-beam LEO systems [2]. With multi-beam payloads, a handover is either intra-satellite (between two beams of one satellite) or inter-satellite (between satellites) [2].

Learning-based handover has been studied to reduce access delay and collisions [1] and as multi-objective decision-making [3]. The Multi-Objective Deep Q-Network (MODQN) of Sun *et al.* casts LEO handover as multi-objective reinforcement learning over throughput, handover cost, and load balancing [2], building on modular multi-objective value decomposition [4]. On the energy side, energy-aware Q-learning has been applied to handover [5], and joint handover and fast beam switching couples energy efficiency (EE) with handover cost [6], with channel and signal-to-interference-plus-noise ratio (SINR) models following 3GPP non-terrestrial-network assumptions and elevation-dependent gain [7–9]. State-augmented reinforcement learning couples assignment or constraints with augmented states [10, 11], and sequential satellite assignment pairs learned values with a coordinated one-to-one matching [12]; these lines assign an agent to a user, satellite, or beam, whereas here the agents are per objective and the allocation is a budgeted many-to-one assignment under per-satellite activation limits. The catfish effect has been used in deep reinforcement learning as an auxiliary agent that stimulates the main learner's exploration, originally in a single-objective energy-saving control task [13]; this paper extends the training idea to one catfish per objective.

MODQN is adopted here as both baseline and backbone. Two limitations motivate this work. First, its raw-throughput reward does not reflect how off-axis angle, antenna gain, interference, and transmit power jointly determine energy efficiency. Second, each user independently selects its highest-scoring beam, and users observing similar states select the same one; under a per-satellite limit on simultaneously active beams this per-user argmax rule can cause beam over-concentration, pushing out the users that do not fit the capacity.

To address both limitations, a Multi-Catfish Coordinated Reinforcement Learning (MCCRL) framework is proposed. The contributions are as follows.

- Angle-aware EE reward. The throughput reward is replaced with an angle-aware energy-efficiency objective that couples the delivered rate (shaped by off-axis angle and interference) to the load-dependent transmit power, rather than ignoring the power cost altogether [6, 7].
- Coordinated valuation–allocation framework. Each objective is treated as a catfish role [13]; the three roles produce a multi-objective combined value for every candidate beam, and a coordinated beam allocation replaces the per-user argmax, jointly deciding which beams to activate

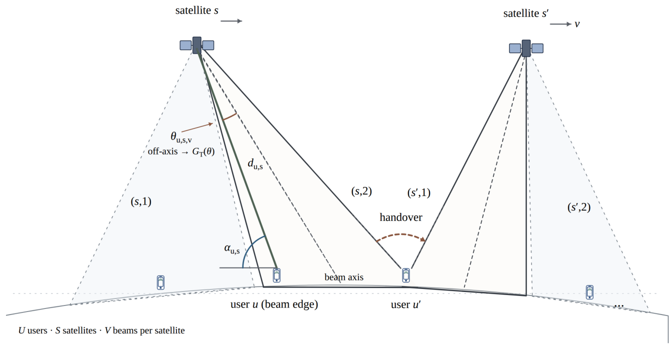

Fig. 1. Multi-beam LEO satellite system model.

and which beam each user connects to under the per-satellite capacity limit. The allocation is a submodular-greedy (facility-location) round [14, 15].

- Attribution and capacity diagnosis. A factorial ablation and a cross-over test (the same trained Q-networks re-decoded under the other selection rule) attribute the de-concentration to the coordinated allocation step; the framework is evaluated against plain MODQN and a set of rule-based static and learned baselines on the normalized weighted metric, and the advantage is located on the efficiency–equity axis.

II. System Model and Problem Formulation

A window of S satellites serves U ground users; each satellite forms up to V directional beams and may keep at most $v_{\max}$ of them active per time step, a per-satellite beam-activation budget, as in dynamic beam management [16]. A binary connection variable $x_{u,s,v}(t) \in \{0,1\}$ marks whether user u connects to beam (s,v), and a binary activation variable $z_{s,v}(t)$ marks whether that beam is active, with $x_{u,s,v} \le z_{s,v}$ and $\sum_v z_{s,v}(t) \le v_{\max}$; $U_{s,v}(t) = \sum_u x_{u,s,v}(t)$ counts the users served by beam (s,v).

a) Channel and SINR: As illustrated in Fig. 1, the per-link gain $h_{u,s,v}(t) = G^{\mathrm{T}}(\theta_{u,s,v}(t))\, G_u^{\mathrm{R}}\, H_{u,s}(t)\, g_{u,s,v}(t)$ is the product of the angle-dependent transmit gain $G^{\mathrm{T}}(\theta)$, a fixed receive gain G_u^{R}, the large-scale attenuation $H_{u,s}(t) = 10^{-L_{u,s}(t)/10}$ (3GPP free-space, shadowing, clutter [7]), and small-scale fading $g_{u,s,v}(t)$. The transmit gain uses a Bessel pattern in the off-axis angle $\theta_{u,s,v}(t)$, the angle between the beam-center direction and the satellite-to-user direction [6, 8],

$$G^{\mathrm{T}}(\theta_{u,s,v}(t)) = G_0 \left[\frac{J_1(\mu)}{2\mu} + 36\, \frac{J_3(\mu)}{\mu^3} \right]^2, \qquad (1)$$

where G_0 is the boresight gain, J_1, J_3 are first-kind Bessel functions of orders 1 and 3, the Bessel-pattern argument $\mu = 2.07 \sin\theta_{u,s,v}(t)/\sin\theta_{3\mathrm{dB}}$, and $\theta_{3\mathrm{dB}}$ is the half-power beamwidth: a larger off-axis angle gives a smaller gain. The cell grid uses a three-color frequency reuse, so a link receives interference only from co-channel active beams: the SINR $\gamma_{u,s,v}(t)$ of a candidate link sums that interference over $\mathcal{I}_{s,v}(t)$, the set of other active beams sharing the color of (s,v), and the achievable rate is the Shannon capacity of the per-color band:

$$\gamma_{u,s,v}(t) = \frac{p_{s,v}(t)\, h_{u,s,v}(t)}{\sigma^2 + \displaystyle\sum_{(s',v') \in \mathcal{I}_{s,v}(t)} z_{s',v'}(t)\, p_{s',v'}(t)\, h_{u,s',v'}(t)}, \qquad (2)$$

$$R_{u,s,v}(t) = \frac{B_c}{U_{s,v}(t)} \log_2\bigl(1 + \gamma_{u,s,v}(t)\bigr), \qquad (3)$$

where $B_c = B_{\mathrm{sys}}/3$ is the per-color bandwidth under the three-color reuse, split equally among the $U_{s,v}(t)$ users on beam (s,v) (the equal bandwidth-sharing rule of [2, 6]), and $\sigma^2 = N_0 B_c F_{\mathrm{NF}}$ is the noise power over that band with noise figure F_{NF}. The served rate is $R_u(t) = \sum_{s,v} x_{u,s,v}(t)\, R_{u,s,v}(t)$.

b) Angle-aware EE reward: The transmit power of a beam is load-dependent,

$$p_{s,v}(t) = z_{s,v}(t) \min\bigl(P_{\mathrm{base}} + P_{\mathrm{scale}}\, U_{s,v}(t)^{\nu},\ P_{\mathrm{beam,max}}\bigr), \qquad (4)$$

with base power P_{base} and coefficients P_{scale}, $\nu \in (0,1)$ setting the concave load growth (our modeling assumption), capped per beam at $P_{\mathrm{beam,max}}$ and per satellite by a total budget $\sum_v p_{s,v}(t) \le P_{\mathrm{sat,max}}$ [6]. The first reward term is the angle-aware energy efficiency

$$r_{1,u}(t) = \eta_u^{\mathrm{EE}}(t) = R_u(t)/p_u^{\mathrm{alloc}}(t), \qquad (5)$$

the delivered rate per unit of allocated power (bit/s/W), where $p_u^{\mathrm{alloc}}(t) = p_{s,v}(t)/U_{s,v}(t)$ is the power given to user u at its served beam (s,v); an unserved user ($\sum_{s,v} x_{u,s,v}(t) = 0$) contributes zero to every objective. Since the bandwidth and power shares cancel, $\eta_u^{\mathrm{EE}}(t) = B_c \log_2(1 + \gamma_{u,s,v}(t))/p_{s,v}(t)$ at the served beam, of order 10^{12} bit/s/W (model-internal, used relatively): the numerator scales with the transmit gain, $\gamma_{u,s,v} \propto G^{\mathrm{T}}(\theta)$ (decreasing in θ), and falls with interference, while the denominator power (4) is load-driven and angle-independent, so at equal allocated power a beam-center user attains a higher rate and energy efficiency than a beam-edge user. Unlike the system-level EE of [6] (total rate over total power), this is a per-user ratio used as a reward.

c) Handover-cost reward: The second objective is the handover cost. Let $\rho_u(t)$ and $\delta_u(t)$ denote the satellite and beam serving user u at time t (from $x_{u,s,v}(t)$); an intra-satellite beam switch costs φ_1 and an inter-satellite switch costs φ_2 ($0 < \varphi_1 < \varphi_2$), and the reward is $r_{2,u}(t) = -\Psi_u(t)$ [2],

$$\Psi_u(t) = \begin{cases} 0, & \rho_u(t) = \rho_u(t-1),\ \delta_u(t) = \delta_u(t-1), \\ \varphi_1, & \rho_u(t) = \rho_u(t-1),\ \delta_u(t) \ne \delta_u(t-1), \\ \varphi_2, & \rho_u(t) \ne \rho_u(t-1). \end{cases} \qquad (6)$$

d) Load-balancing reward: The third objective is load balancing, the negative spread between the busiest and least-busy active beam [17],

$$r_{3,u}(t) = -\frac{1}{U} \begin{pmatrix} \displaystyle\max_{(s,v)\in\mathcal{B}(t)} \tilde{R}_{s,v}(t) - \\ \displaystyle\min_{(s,v)\in\mathcal{B}(t)} \tilde{R}_{s,v}(t) \end{pmatrix}, \qquad (7)$$

where $\mathcal{B}(t)$ is the set of active beams (an inactive beam carries no load) and $\tilde{R}_{s,v}(t) = \sum_u x_{u,s,v}(t)\, R_{u,s,v}(t)$ is the aggregate load of beam (s, v).

Following the multi-objective formalization of MODQN [2], the long-term handover problem is stated as a vector optimization over the connection and activation variables, rather than as a single weighted sum:

$$\begin{aligned} \max_{\pi} \;\; & \mathcal{J}(\pi) = \big(\mathcal{J}_1(\pi),\ \mathcal{J}_2(\pi),\ \mathcal{J}_3(\pi)\big) \\ \text{s.t.} \;\; & C_1:\ x_{u,s,v}(t), z_{s,v}(t) \in \{0,1\},\ \textstyle\sum_{s,v} x_{u,s,v}(t) \le 1, \\ & C_2:\ x_{u,s,v}(t) \le z_{s,v}(t),\ \textstyle\sum_v z_{s,v}(t) \le v_{\max}, \end{aligned} \tag{8}$$

where $\mathcal{J}_j(\pi) = \mathbb{E}_\pi\big[\sum_{t=0}^{T-1} \bar{r}_j(t)\big]$ is the expected long-term return of objective j over a horizon of T steps, $\bar{r}_j(t)$ is the user mean of the per-user rewards of (5)–(7), and the policy π selects the joint action $a(t) = \{x_{u,s,v}(t), z_{s,v}(t)\}$: $\mathcal{J}_1$ maximizes the long-term angle-aware energy efficiency, $\mathcal{J}_2$ minimizes the handover cost, and $\mathcal{J}_3$ minimizes the load imbalance. C1 makes each user attach to at most one beam, and C2 makes it attach only to an active beam and caps each satellite at $v_{\max}$ active beams, a beam-activation budget that MODQN does not impose. The conflicting rewards are kept as a vector rather than summed [2, 4].

For evaluation, the three objectives are combined into one normalized score

$$\mathcal{J}_w = \sum_{j=1}^{3} \omega_j\, \frac{\bar{r}_j}{c_j}, \tag{9}$$

where $\bar{r}_j$ is the average of the per-user reward $r_{j,u}(t)$ over all users and the whole horizon, c_j is a fixed per-objective normalization constant that brings the three otherwise incommensurable magnitudes to a comparable scale, and ω_j are preference weights shared with the scalarized policies below (both are listed in Table I). All compared methods share the same ω_j and c_j. $\mathcal{J}_w$ is the headline scalar metric (the weights follow [2]). Because the motivating failure mode is tail starvation, coverage metrics (minimum per-user coverage and served fraction, Section IV) are reported as co-primary criteria. The per-objective components are reported separately (Table II and the per-objective breakdown in Section IV), so the weighted score hides no per-axis regression.

III. The MCCRL Scheme

The backbone trains one Q-network per objective and forms a linear scalarization $\sum_k \omega_k Q_k$; the original method then lets each user pick its own highest-value beam (per-user argmax) [2, 18, 19]. This per-user rule has no mechanism to coordinate how many beams a satellite activates in total, which is the proximate origin of over-concentration under a tight $v_{\max}$.

a) Per-objective catfish training: Following the catfish effect [13], each objective is assigned a catfish role—a per-objective Q-network trained with two catfish mechanisms: a value-stratified experience replay that oversamples the rare high-return transitions where users compete for the same

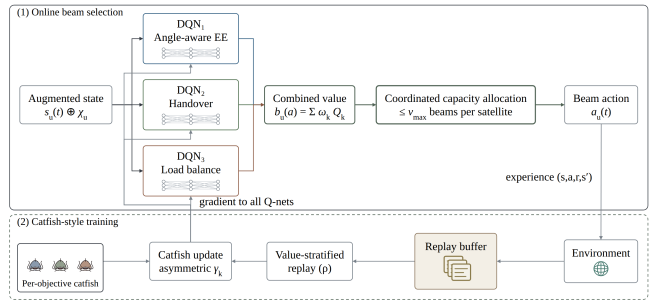

Fig. 2. The proposed MCCRL framework: per-objective catfish valuation followed by coordinated beam allocation.

beam [20], and an asymmetric per-objective discount β_k that gives the handover objective a shorter effective horizon than the other two. Each Q-network observes a user state that stacks the connection status, candidate-link SINRs and off-axis angles, and the beam-load distribution; a congestion-context state augmentation χ_u (historical occupancy, number of candidate competitors, signal rank) is concatenated to form the augmented state $\tilde{s}_u(t)$.

b) Coordinated beam allocation: The online decision is reorganized into two steps (Fig. 2). In valuation, the three catfish produce, for every user u and every candidate beam $a = (s, v)$, a multi-objective combined value $b_u(a)$ (user u's score for beam a),

$$b_u(a) = \sum_{k=1}^{3} \omega_k\, Q_k(\tilde{s}_u(t), a). \tag{10}$$

Per user these scores are shifted so the worst feasible candidate is zero, $\hat{b}_u(a) = b_u(a) - \min_{a' \in \mathcal{A}_u(t)} b_u(a')$, making the round invariant to per-user value offsets, where $\mathcal{A}_u(t)$ is the set of beams reachable by user u at time t.

In allocation, a coordinated greedy round replaces the per-user argmax. Starting from an empty opened set $\mathcal{S}$, it repeatedly opens the beam that adds the most total value. Let $o(u) = \max\{0, \max_{a \in \mathcal{S} \cap \mathcal{A}_u(t)} \hat{b}_u(a)\}$ be the best value user u already obtains from the opened set (zero if none of its candidates is open); writing $[x]^+ = \max(x, 0)$, a not-yet-opened beam (s, v) on a satellite still under budget has marginal gain

$$\Delta(s, v) = \sum_{u:\,(s,v) \in \mathcal{A}_u(t)} \big[\,\hat{b}_u(s, v) - o(u)\,\big]^+. \tag{11}$$

The round opens $(s, v)^\star = \arg\max_{(s,v)} \Delta(s, v)$ and repeats until every per-satellite budget ($\sum_v z_{s,v} \le v_{\max}$) is spent or no positive gain remains. Each user then attaches to its best opened candidate; a user with no opened candidate is bumped—unserved that step, carrying no power and contributing zero to every objective—so the executed allocation never violates C1–C2.

This round is the classical facility-location assignment [14, 15]: with $F(\mathcal{S}) = \sum_u \max\{0, \max_{(s,v) \in \mathcal{S} \cap \mathcal{A}_u(t)} \hat{b}_u(s, v)\}$ the total assigned value of an opened set (a user reached by no open beam contributes zero), $\Delta(s, v)$ equals the marginal increase $F(\mathcal{S} \cup \{(s, v)\}) - F(\mathcal{S})$; F is monotone submodular and the per-satellite budgets form a partition matroid, so the greedy round is $1/2$-approximate for F [21]—a guarantee on the learned values at the current step, not on the long-horizon objective (8).

It costs $O(S^2 V v_{\max} U)$ per step ($\sim 3.4 \times 10^4$ operations; lazy-greedy pruning applies) and, unlike the per-user argmax, is computed centrally from the candidate values (a coordination assumption shared by the fixed-rule allocation of Section IV), so it accounts for the total number of active beams and for competition across users, and is designed to counteract the over-concentration onto a few beams; the realized de-concentration is measured in Section IV. The framework integrates this coordinated allocation into the MODQN backbone; it is not a new value-decomposition algorithm.

c) *Training procedure:* The three Q-networks are trained jointly with Double DQN [18, 19]. At each step the environment returns the three rewards r_1, r_2, r_3, and each Q_k is updated by a one-step temporal-difference target under its own discount β_k (with differing discounts the combined value b_u is a scalarization of per-objective values, not the value function of a single discounted problem); the next-step action inside the target is recomputed with the same coordinated allocation rule, so the training bootstrap matches the deployed decision rule. The value-stratified replay marks a transition τ as critical when its normalized weighted reward $\mathcal{J}_w(\tau) = \sum_j \omega_j \, r_j(\tau)/c_j$ (the per-transition analogue of (9)) exceeds a running statistic,

$$\mathcal{J}_w(\tau) > m_J + \kappa \, s_J, \tag{12}$$

where m_J and s_J are the running mean and standard deviation of that reward, maintained online over past transitions, and κ is a threshold multiplier; critical transitions enter a priority subset, and every mini-batch draws a fixed fraction f_r from this subset and the remainder uniformly, so the rare high-competition states are revisited more often. The general-training variant sets $f_r = 0$, recovering uniform sampling. Target networks are synchronized periodically and the exploration rate is annealed over training; all hyper-parameter values are listed in Table I. At inference the trained Q-networks are combined into $b_u(a)$ and decoded by the coordinated allocation above.

IV. Experimental Results

The environment reuses the multi-beam LEO system of Section II: a window of $S = 4$ satellites, each with $V = 7$ beam slots (28 actions), serving $U = 100$ users, with $v_{\max} = 3$ active beams per satellite in the main experiment, a deliberately tight budget. Table I lists the physical and training constants. Each objective trains one Q-network with Double DQN for 3000 episodes (fewer than the backbone's 9000 [2], a budget the congestion-augmented state and tight per-window action space make sufficient here); the five MODQN and MCCRL variants all train on the reward vector of Section II under the same optimizer and schedule, so the comparisons

TABLE I
Simulation and training parameters.

Parameter	Value	Parameter	Value
Altitude; carrier	780 km; 20 GHz	Constellation	Walker-δ 180/9/1
Area; user speed	200×90 km; 30 km/h	Slot; episode	1 s; 10 slots
B_{sys} (3-color reuse)	500 MHz	N_0; noise fig.	-174 dBm/Hz; 1.2 dB
G_0; $\theta_{3\text{dB}}$	40 dBi; 3.32°	RX gain; QoS floor	35 dBi; 10 Mbit/s
P_{base}; P_{scale}; ν	0.25 W; 0.35 W; 0.5	Power caps (beam; sat.)	10 W; $10^{1.3}$ W
φ_1; φ_2	0.5; 1.0	ω	$(0.5, 0.3, 0.2)$
c_j	$2.34{\cdot}10^{15}$; 300.3; $6.13{\cdot}10^{9}$	β_k (catfish; general)	$(0.99, 0.90, 0.99)$; 0.9
κ; f_r	0.5; 0.25	Episodes; batch; LR (Adam)	3000; 128; 0.01
Hidden layers	100–50–50 (tanh)	Replay; sync; ε	50000; 50 ep.; 1.0→0.01/2000

isolate the decision rule and the training mechanisms rather than the reward design. The catfish variants use the value-stratified replay and the asymmetric per-objective discounts; the general-training variants use uniform sampling ($f_r = 0$) and a symmetric discount. The congestion augmentation χ_u enlarges the state dimension from 140 to 224 for every trained variant except plain MODQN. The five learned variants use three seeds evaluated on 48 matched episodes; intervals are 95% percentile bootstrap over the 48 seed-averaged episodes (20000 resamples; episode-paired for the ablation deltas). The two single-objective DQNs are trained on five independent seeds under a separate field protocol (DQN-throughput on raw throughput, DQN-scalar on a positive-affine transform of the weighted score (9)) and report five-seed t-intervals, so their rows are cross-protocol context rather than a matched comparison. The static baselines (RSS-max, round-robin, and a fixed-rule coordinated allocation that decodes observed signal as the value, without learning) use the same matched 48-episode protocol.

a) *Over-concentration of plain MODQN:* At $v_{\max} = 3$ it and the variant that only adds the congestion state but keeps the per-user argmax (MODQN + aug) both over-concentrate: the mean number of active beams is about 3 of a possible 12, the served fraction (the fraction of users meeting their QoS target) is about 0.29, and the minimum coverage (the served-time fraction of the worst-off user) is 0. This follows from the per-user argmax rule: users with similar states select the same beam, its load exceeds capacity, and users beyond the budget are excluded. That the selection rule, rather than insufficient training, is the proximate lever is confirmed by the cross-over test below. Whether this over-concentration is inherent to the shared-Q architecture or specific to the capacity-constrained regime is left open.

b) *Framework versus baselines:* Table II compares all methods at $v_{\max} = 3$.

The proposed MCCRL and its inference-only variant (coordinated allocation without catfish training) reach $\mathcal{J}_w \approx 4.8$–$5.0 \times 10^{-4}$, far above plain MODQN ($\approx -1.0 \times 10^{-5}$), with non-overlapping confidence intervals. Over-concentration is mitigated: the effective number of active beams rises from about 3 to about 9–11, the served fraction from 0.29 to 0.97–0.99, and the minimum coverage from 0 to 0.999. The coordinated MCCRL variants also exceed the rule-based static baselines RSS-max (1.55×10^{-4}) and round-robin (1.60×10^{-4}) with separated intervals. Round-robin opens many beams yet still has zero minimum coverage: opening beams is not enough without proper assignment. Conversely, the fixed rule reaches full coverage with only 3.6, so the active-beam count is a de-

TABLE II
COMPARISON OF ALL METHODS AT $v_{\max} = 3$.

Method	$\mathcal{J}_w$	95% CI	EE	Min-cov	Served	Beams
MODQN (plain)	-0.10	$[-0.54, 0.37]$	1.01	0.00	0.29	3.0
MODQN + aug	-0.16	$[-0.61, 0.33]$	1.02	0.00	0.29	3.0
MODQN + catfish	$+0.45$	$[-0.04, 0.98]$	1.19	0.00	0.30	3.0
RSS-max	$+1.55$	$[1.01, 2.11]$	1.50	0.00	0.31	3.2
Round-robin	$+1.60$	$[1.27, 1.96]$	2.23	0.00	0.41	11.7
DQN-throughput	$+3.54$	$[1.73, 5.36]$	3.20	0.00	0.63	$5.6^{\dagger}$
Coordinated Alloc.	$+4.44$	$[3.53, 5.30]$	3.73	1.00	0.95	3.6
MCCRL (Proposed)	$+4.82$	$[4.53, 5.11]$	3.85	0.999	0.97	9.4
MCCRL (Inference-Only)	$+4.97$	$[4.67, 5.27]$	3.92	0.999	0.99	10.6
DQN-scalar	$+6.41$	$[5.96, 6.87]$	4.79	0.00	0.89	$12.8^{\dagger}$

$\mathcal{J}_w$ and its CI in 10^{-4}; EE is the angle-aware $\bar{r}_1$ in 10^{12} bit/s/W at the operating point. Intervals: episode bootstrap for the three-seed MODQN/MCCRL and static rows, five-seed t-intervals for the DQN rows (Setup). A minimum coverage of 0 means the worst-off user is never served. MCCRL (Inference-Only) is fully trained (coordinated TD targets); only catfish is off. †DQN rows: distinct requested beams (requests beyond the 12-beam budget are bumped, carrying no power); other rows: concurrently loaded beams (at most 12).

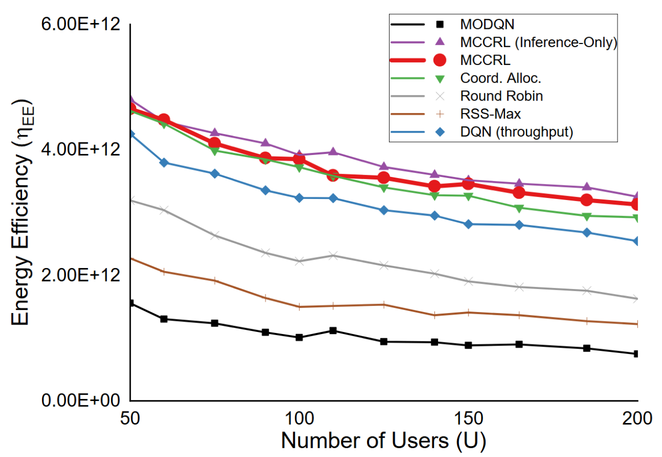

Fig. 3. Energy efficiency (angle-aware r_1, 10^{12} bit/s/W) versus the number of users. DQN-scalar reaches higher EE but under-serves the tail (min-cov 0 at the Table II operating point, ≤ 0.85 over the sweeps) and is omitted.

concentration diagnostic rather than a merit metric, and more active beams draw more base power (4).

c) Efficiency versus equity: The fixed-rule coordinated allocation reaches $\mathcal{J}_w = 4.44 \times 10^{-4}$ and minimum coverage 1.0; the inference-only MCCRL's margin over the fixed rule has a confidence interval covering zero, so the extra benefit of learning over the fixed rule is not resolved at this seed count; the de-concentration is attributable mainly to the coordinated allocation step, not to learning. Conversely, DQN-scalar attains the highest weighted score (6.41×10^{-4}, above the inference-only MCCRL's 4.97×10^{-4}); no claim is made of beating DQN-scalar on the weighted scalar. However, DQN-scalar reaches that score by starving the tail: its minimum coverage is 0 and its Jain fairness index over the users' cumulative episode throughput [22] is about 0.69, whereas the inference-only MCCRL serves almost every user. The proposed framework is therefore positioned on the efficiency–equity axis — the only learned method that simultaneously attains high energy efficiency and near-full coverage (0.999) at tight capacity — rather than as a maximizer of the weighted scalar alone.

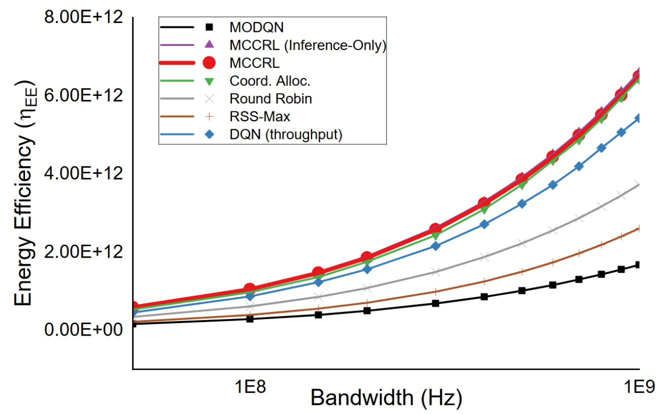

Fig. 4. Energy efficiency versus the available bandwidth; DQN-scalar omitted as in Fig. 3.

TABLE III
CROSS-OVER DECODE TEST AT $v_{\max} = 3$ (MATCHED 48-EPISODE PROTOCOL): THE SAME TRAINED Q-NETWORKS, ONLY THE DECODE SWAPPED. CELLS: $\mathcal{J}_w$ (10^{-4}) / MIN-COV; DIAGONALS ARE THE NATIVE VARIANTS OF TABLE II).

Trained Q-networks	Argmax decode	Coordinated decode
From MODQN + aug	-0.16 / 0.000	$+4.76$ / 1.000
From MCCRL (Inference-Only)	$+0.19$ / 0.000	$+4.97$ / 0.999

d) Per-objective components: On the handover axis the trained coordinated variants pay the lowest cost ($\bar{r}_2 = -0.088$ to -0.090, intervals separated from plain MODQN's -0.100 and the fixed rule's -0.189; the fixed rule reassigns about twice as often, the one axis where learning improves on it). The load-spread term regresses ($\bar{r}_3 \approx -7.7 \times 10^6$ versus -3.8×10^6 for plain MODQN), as spreading load across ten beams widens the busiest-to-least-busy gap; the weighted score absorbs this disclosed regression.

e) Ablation: The framework changes two things: the selection rule and the catfish training. A 2×2 ablation separates them; the four cells share the χ_u-augmented state and differ only in the selection rule (used both online and inside the TD target) and the catfish training of Section III. Replacing per-user argmax with coordinated allocation moves $\mathcal{J}_w$ from -1.6×10^{-5} to 4.97×10^{-4} and minimum coverage from 0 to 0.999. Under the per-user argmax, catfish training does shift $\mathcal{J}_w$ by $+6.1 \times 10^{-5}$ (95% CI $[+5.0, +7.3] \times 10^{-5}$; MODQN + aug versus MODQN + catfish) but leaves the collapse intact (minimum coverage 0): a real training-side effect that does not substitute for coordination. Adding catfish training under coordinated allocation changes $\mathcal{J}_w$ by only -1.5×10^{-5}, whose confidence interval covers zero: no significant additional gain in this regime, plausibly because coordination removes the beam-competition collisions that the value-stratified replay is designed to oversample. A cross-over test makes the attribution direct (Table III): taking an argmax-trained Q and decoding it with coordinated allocation lifts minimum coverage from 0 to 1.000 (with $\mathcal{J}_w$ rising to 4.76×10^{-4} and the effective beams from 3 to 10.3), while decoding a coordinated-trained Q with per-user argmax collapses it back (coverage 0, $\mathcal{J}_w = 1.9 \times 10^{-5}$). The same

Q-values flip between collapse and full coverage when only the decode rule changes, so the de-concentration is driven by the coordinated allocation step; the evidence therefore supports neither crediting the catfish training with the improvement nor the claim that removing it causes collapse.

f) Sensitivity: Figs. 3 and 4 re-evaluate the fixed trained models while sweeping the number of users and the available bandwidth. Among the methods that keep near-full coverage, the coordinated MCCRL variants lead the energy efficiency point-wise across both sweeps: the margins over plain MODQN, RSS-max, and round-robin are interval-separated throughout; against the fixed rule they separate at most user counts and at low bandwidth and overlap elsewhere. DQN-scalar reaches a higher raw EE, but only by leaving the tail unserved: its minimum coverage stays between 0.1 and 0.85 over the sweep and is 0 at the tight $v_{\max} = 3$ point of Table II. The weighted-scalar advantage over plain MODQN is largest when capacity is tight and shrinks as $v_{\max}$ grows and over-concentration becomes mild to begin with (matched $v_{\max}$ sweep, not shown).

V. Conclusion

This paper has replaced the throughput reward of MODQN with an angle-aware energy-efficiency objective and reorganized the online decision into a valuation step built on three per-objective catfish roles followed by a coordinated beam allocation replacing the per-user argmax. Under a tight per-satellite beam budget the proposed MCCRL framework substantially mitigates the over-concentration of plain MODQN, raising the number of served users and active beams and beating it and the rule-based static baselines on the normalized weighted metric, while remaining the only learned method that simultaneously attains high energy efficiency and near-full coverage. Ablation attributes the de-concentration to the coordinated allocation step rather than to the catfish training, and the advantage is located on the efficiency–equity axis rather than on the weighted scalar, on which a simpler scalar-trained DQN scores higher at the cost of zero minimum coverage. Whether the over-concentration is rooted in the environment, the state, or the shared-Q algorithm remains open; integrating the competitive-reward mechanism of the source catfish method and testing across more environments are left for future work.

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